Temporal Causality: Modeling Time as the Agent of Quantum Collapse

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0. Background

In Temporal Primacy: Time as Foundational Force in Quantum Collapse and Emergence (2025), the hypothesis
was introduced that time is not merely a passive dimension but an active, causal force driving quantum wave function collapse. The earlier paper developed a conceptual framework for how temporal flow might generate temporal tension-an internal pressure within probabilistic systems-leading to collapse when a threshold is reached. The present work builds directly on that foundation. While Temporal Primacy outlined the philosophical
and theoretical motivation for the hypothesis, the current paper focuses on turning the concept into a formal model with clear definitions, equations, and experimental proposals. This shift from broad theory to specific modeling is intended to make the hypothesis easier to test, refine, or falsify.

1. Introduction

This paper builds directly on the Temporal Primacy Hypothesis proposed in earlier work, which framed time as an
active, foundational force responsible for quantum wave function collapse, the emergence of order, and potentially the rise of consciousness. Here, the focus shifts from broad conceptual framing to specific modeling.
The aim is to describe, in clear mathematical and physical terms, how the flow of time might exert a measurable
influence on collapse probability - and to outline scenarios where these effects could be tested. If Temporal Primacy asked 'Could time be the agent?', this paper asks 'What would that agency look like in equations, experiments, and predictions?'

2. Core Hypothesis

We start with the idea that temporal tension - the cumulative 'pressure' of time passing - can push indeterminate systems toward collapse. This tension, ?(t), increases until it meets or exceeds a collapse

threshold ?_p. At that point, a specific outcome is selected from the probability distribution. We also define a temporal agency function, A(t), which scales the influence of time based on environmental or system conditions
- such as entropy growth, gravitational time dilation, or information density. Hypothesis Statement: Collapse is
initiated when ?(t) · A(t) reaches a critical value. Both ?(t) and A(t) can vary depending on the system's physical
and informational context.

3. Mathematical Model

We adapt the collapse model from Temporal Primacy into a form that can handle variable thresholds and environmental dependencies: d?/dt = -A(t) · ?(t) · ? Where: - ? is the system's quantum state. - ?(t) is temporal tension, modeled as a non-negative, increasing function. - A(t) modulates how strongly time acts on the system. -
Collapse occurs when ?(t) ? ?_p. Possible forms of ?(t) include: 1. Linear Growth: ?(t) = k · t - simplest case, constant rate of temporal tension build-up. 2. Entropy-Driven: ?(t) = k · S(t) - tension grows with system entropy.
3. Curvature-Dependent: ?(t) = k · f(R) - tension increases in curved spacetime. Possible forms of A(t) include: 1. Constant Agency: A(t) = a_0 - uniform influence of time. 2. Local Condition Scaling: A(t) = a_0 (1 + ? E(t)) -
increased influence in high-energy or high-information states.

4. Experimental Designs

4.1 Quantum Optics: Time-Dependent Collapse Rates Set up a double-slit or Mach-Zehnder interferometer with
the ability to delay measurement. Vary only the time interval between state preparation and measurement, while
holding environmental noise constant. Prediction: if collapse probability is partly time-driven, longer delays should
subtly change interference visibility even without added decoherence.
4.2 Entanglement Under Temporal Gradient Prepare entangled pairs and place one qubit in a region of slightly different gravitational potential (e.g., mountaintop vs. valley) to induce gravitational time dilation. Prediction: small but measurable drift in correlation

statistics over time as local ?(t) diverges.

4.3 Artificial Temporal Threshold Simulations In a virtual environment,
program agents or decision systems to 'collapse' once modeled ?(t) exceeds a threshold ?_p. Track patterns in collapse timing, symmetry, and emergent complexity. This could be run on classical or quantum simulators. Prediction: collapse timing will show statistical clustering around predictable temporal tension peaks.

5. Implications

If time can drive collapse: - For Quantum Mechanics: This reframes the measurement problem by making collapse an intrinsic temporal process rather than requiring an external observer or purely random trigger. - For
Relativity: Local variations in time flow could directly influence quantum outcomes, linking general relativity and
quantum mechanics in a new way. - For Consciousness Research: If decision-making or awareness arises from collapse events, then temporal flow - not computation alone - may be a fundamental substrate for consciousness. - For Technology: Quantum computers, if sensitive to ?(t), may need to account for local temporal
gradients in error correction and qubit lifetime management.

6. Conclusion

The Temporal Causality model gives shape to the idea first posed in Temporal Primacy: that time is not merely a
backdrop for events, but an active agent in selecting them. By defining temporal tension ?(t), temporal agency A(t), and collapse thresholds ?_p, we move from speculation toward something testable. If supported by experiment, this framework could unify aspects of quantum mechanics, relativity, and consciousness research -
and if disproven, it will still help clarify why collapse happens when it does. Either way, the hypothesis deserves to
be tested.

Appendix A: Example Functions for ?(t) and A(t)

1. Example Temporal Tension Growth: - Linear: ?(t) = k · t (steady build-up over time) - Exponential: ?(t) = k · e^(?
t) (rapid build-up in unstable systems) - Entropy-Linked: ?(t) = k · log(S(t) + 1) (slows down in low-entropy

environments)

2. Example Temporal Agency Scaling: - Constant: A(t) = a_0 (time acts uniformly) - Curvature-Responsive: A(t) = a_0 (1 + ? R) (stronger effect in curved spacetime) - Energy Density-Responsive:

A(t) = a_0 (1 + ? E(t))

Appendix B: Additional Simulation Scenario

Nested Threshold Simulation: Instead of a single collapse threshold ?_p, model systems with multiple nested thresholds - analogous to complex decision-making processes or layered quantum systems. Each layer collapses when its own ?_{p,i} is met, but lower layers feed back into higher ones, altering their tension rates. This could be implemented in computational models to see whether hierarchical collapse produces emergent coherence patterns similar to those in biological systems.